CS4050 - Lab 4 - Ashton Wooster

No AI tools were used for this assignment. I used the previous memoization lab with text justification for guidance.

Exercise 1:

Key Signal: 5 2 3 2 2 1 4
Recorded Signal: 3 2 2 1

I wrote a brief code snippet to calculate this. Below is the output:

SHIFT 0:
Score: 1
SHIFT 1:
Score: 1
SHIFT 2:
Score: 4
SHIFT 3:
Score: 1
The highest score, 4, occurs at shift 2
KEY:
5, 2, 3, 2, 2, 1, 4,
BEST SHIFT, 2, ANSWERS:
-1, -1, 3, 2, 2, 1, -1,

Note that “-1” is the sentinel value indicating no input. The code is included in this directory under “shifts.c”

Exercise 2:

1. Explain why testing all offsets (shifts) guarantees the optimal score.
   This is the equivalent of brute forcing every possible solution. Then, we are checking the entire solution space, and if the solution space is not empty, the optimal score must exist within the solution space. So we are guaranteed to find it.

2. What does Shift Left mean in signal alignment?
   Shift Left means that for a given signal, every value at index *i* gets the value at index *i+1*.

3. What does No Shift mean?
   No Shift means that for a given signal, every value at index *i* remains the same.

4. What does Shift Right mean?
   Shift Right means that for a given signal, every value at index *i* gets the value at index *i-1*.

5. Where else is this algorithm used? provide a real word scenario.
   This algorithm for signal shifting could also be used for aligning audio with a video. You would calculate the fit of the audio track at a given time to the video, perhaps by using a machine learning model, and then shift until you find the optimal shift.

Exercise 3:

1. Submit grade.c
   Provided in the directory.

2. What does the key signal represent in the grading system?
   The key signal represents the solution key for the exam. Each item in the signal corresponds to the correct answer for the question.

3. What does the exam signal represent?
   The exam signal represents the student's exam answer choices. Each item in the signal corresponds to their selected answer.

4. Why might the exam signal be shorter than the key signal?
   The exam signal would be shorter in the cases where the student did not answer all of the questions. In these cases, those answers are omitted, so the signal is shorter.

5. Why must the grading program test multiple shifts?
   The grading program must test multiple shifts to find the best possible shift for the student. This is likely the correct shift. There could be omitted answers throughout the signal, so it must be tested against all possible shifts of all characters.

6. Why must the program check array bounds when comparing signals?
   The program must check array bounds to assure that it does not go out of bounds on checking. For a key of length k, an exam of length e, there are only $k-e+1$ valid solutions, or $k-e$ valid shifts. If we are iteratively solving this, we do not want to go backwards into previously shifted segments or forwards out of the array.

7. Write pseudocode for computing the best alignment score

Algorithm for Best Alignment | Align()

```
If key.length() == 0 or key.length() < student_ans.length(): // base cases
    return 0;
If memo[key.length()][student_ans.length()]: // already computed this scenario
    return memo[key.length()][student_ans.length()];
shift_score = grade_memo(key[1:], student_ans[1:]); // score if student answer skipped
keep_score = grade_memo(key[1:], student_ans); // score if student answer kept
keep_score += (key[0]==student_ans[0]); // grade current question
memo[key.length()][student_ans.length()] = max(keep_score, shift_score); // store highest
return memo[key.length()][student_ans.length()];
```
